

7. **Image 7.1** shows the pathway from a tooth to an area of the brain which generates the sensation of pain. It also shows the site of action of two local anaesthetics used in dentistry. **Image 7.2** shows a reflex arc.

Image 7.1

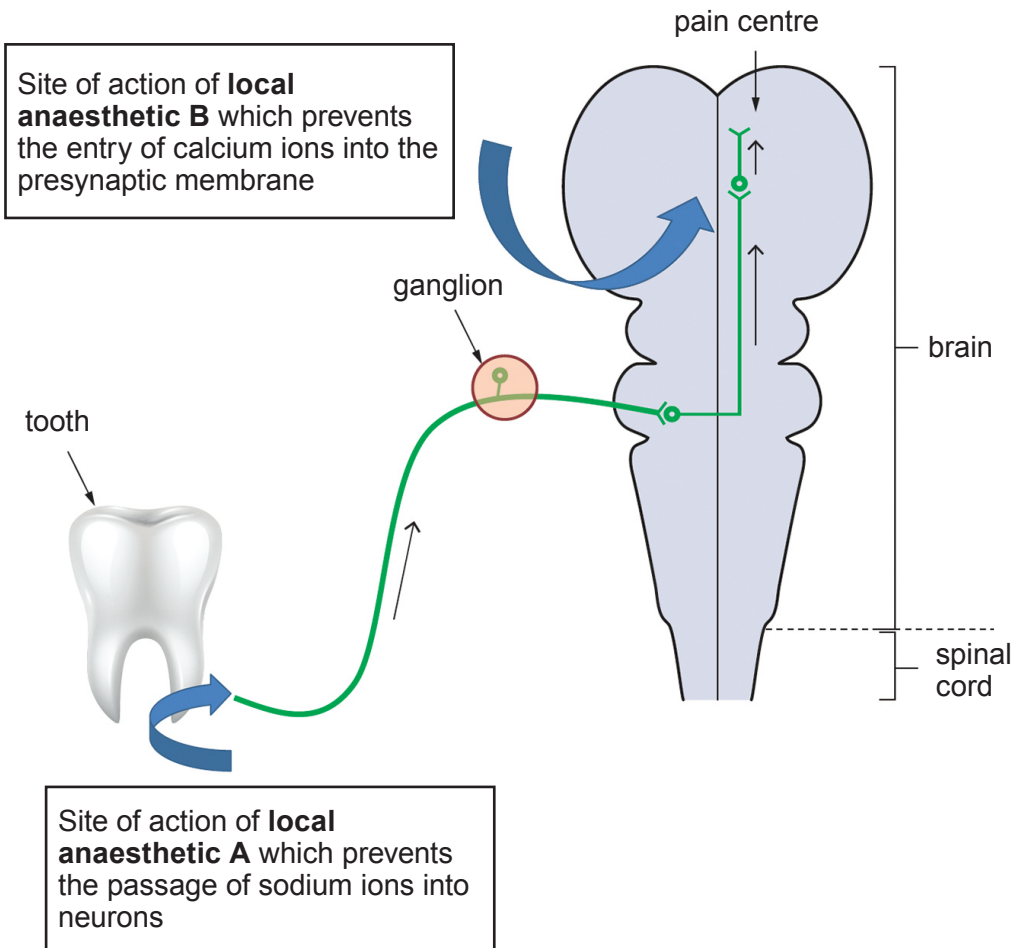
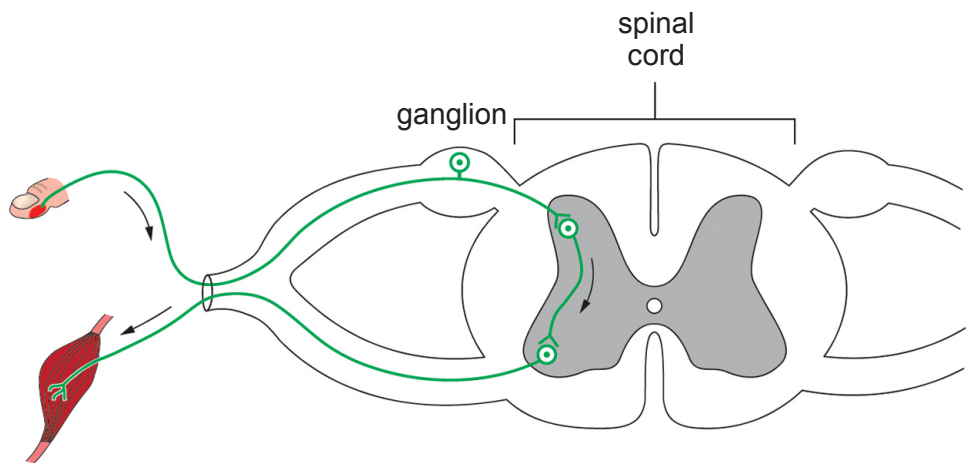


Image 7.2



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Compare and contrast the pathway shown in **Image 7.1** with the reflex arc in **Image 7.2**.

Using your knowledge of the generation of action potentials, suggest how anaesthetic **A** will prevent pain.

Using your knowledge of synaptic transmission, suggest how anaesthetic **B** could also prevent pain. [9 QER]



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